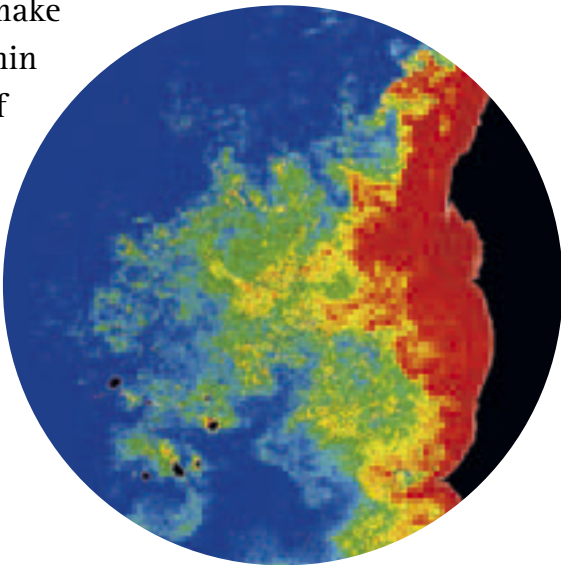




LIFE IN SHALLOW SEAS

▲ SHALLOW FRINGES
The fringes of the continents are covered by sunlit seas that are generally less than 650 ft (200 m) deep. The shallow seafloors are covered with sand, mud, and other sediments that have been swept off the land by rivers and coastal erosion. These sediments are enriched with the remains of land plants and animals, as well as those of dead sea creatures, so they are full of the nutrients needed by marine life.

Most marine life lives in shallow seas, on the continental shelves that fringe the deep oceans. The shallow water makes it easy for essential nutrients from the seabed to reach the sunlit surface zone, where phytoplankton can use them to make food. The seabed itself may also lie within the sunlit zone, giving a wide variety of bottom-dwelling animals direct access to rich supplies of living food.



► SCOURING CURRENTS
Most nutrients that support the flourishing marine life of coastal seas are scoured up from the shallow seabed by storms and currents. This effect is most intense where surface currents moving away from the coast draw deep water up to the surface. This satellite image of the west African coast shows one of these upwelling zones, with red and yellow indicating dense clouds of multiplying phytoplankton.

SHELF LIFE



ON THE BOTTOM
A lot of animals live on the seabed, either buried in the mud and sand or hidden among rocks and wrecks. They include burrowing clams and worms, attached animals such as mussels and sea anemones, and mobile starfish, sea urchins, crabs, crayfish, and lobsters.



NEAR THE SEABED
Many fish such as rays, flatfish, and this Atlantic cod feed on or near the seabed in shallow seas. There is plenty to eat, and prey animals such as crabs and burrowing clams are easy to catch compared with the fast-moving fish that live in the waters above.



OPEN WATER
The zooplankton that swarm in shallow, sunlit, nutrient-rich seas are preyed upon by large shoals of fish that swim in the open water well above the seabed. The fish and their prey move toward the surface at night, and back down into deeper water by day.



SURFACE DIVERS
Many animals that breathe air at the surface are able to reach the shallow seafloor very easily, and feed on the rich variety of prey that lives there. They include a variety of diving seabirds such as auks and cormorants, and mammals such as sea otters and seals.